ALAGAPPA UNIVERSITY, Karaikudi-3

(Reaccredited with 'A' Grade by NAAC)

Ph.:04565 228100(30 lines) Fax:04565 225202

Tender Notification No.2012/WiFi

Sealed Tenders are invited from the eligible suppliers for the supply and installation of WiFi instrument in the campus. The Tenders should deposit an amount of Rs.48,000/= as EMD for the item along with the tender. The rates should be quoted in the price schedule available in the detailed Tender Advt. in our website. The tenders should reach on or before 10.10.2012 at 3.00 pm. The same will be opened at 4.00 pm on the same day before the tender person(s). The tender documents may be downloaded in the university web site: www.alagappauniversity.ac.in with a tender fee of Rs.9,450/= or send a demand draft for Rs.9,450/= in the name of REGISTRAR, Alagappa University payable at Karaikudi. The undersigned reserves the rights to accept or reject or cancel any tender in part or in full without assigning any reasons.

Place: Karaikudi

Date: 21.09.2012 ***REGISTRAR***

ALAGAPPA UNIVERSITY, KARAIKUDI
(Reaccredited with "A" Grade by NAAC)

Ref:A1/2012/WiFi/ Dt: 21.09.12

SEALED TENDERS IN DUPLICATE are invited from reputed firms for the supply and installation of **"WiFi - Equipments"** to this University, so as to reach the THE REGISTRAR, ALAGAPPA UNIVERSITY, KARAIKUDI -630 003 on or before 03.00 p.m. on 10.10.2012 and should be superscribed as follows:

"TENDER REF.NO. Ref:A1/2012/WiFi/ "due on 10.10.2012"

1. The same will be opened by the Registrar or any of his authorized representatives in the presence of such of those tenderers or their duly authorized agents who may be present at that time. The date of opening the tender will be opened on 10.10.12 at 4.00 p.m.

   (i) The prices quoted should be in Indian Rupees and should include sales tax, Central Excise and customs Duty if any, freight to destination and levies and should be for delivery at Alagappa University, Karakudi - 630 003.

   (ii) The tenderers are requested to quote <u>their Sales Tax/VAT Registration Number</u> without fail. In respect of Inver-State sales, Form "D" declaration will be issued wherever the concessional rate of Central Sales Tax is applicable provided always it is hereby agreed and declared that in case when the University has doubt whether the sales tax is at all payable on the transaction in question. The University is authorized to withhold the amount of tax until the party produces an order of the court of competent jurisdiction, declaring the liability of the transaction to "Sales-Tax".

   (iii) Ownership of materials shall rest with the supplier until the same is delivered to this University in good condition.

2. Every Tender should be accompanied by an EMD of **Rs.48,000/=** through Demand Draft drawn in favour of the Registrar, Alagappa University payable at Karaikudi. **The fact of remittance of EMD should be superscribed on the outer cover of the tender. Tender covers without such superscription will not be opened**. The deposit of successful tenderer which will carry no interest will be refunded only after the satisfactory completion of the supply of materials ordered and audit of accounts thereto. Request from firms for waiver of tender deposit will not be considered. No adjustment from the amount due to tenderer will be permitted to be treated as deposit for this tender. ***<u>Tenders without EMD will not</u>***

be considered. Registered small scale industries who provide evidence of existence from competent authorities will be exempted from payment of EMD. In respect of unsuccessful tenders, the EMD will be released immediately after finalization of tender.

(a) 5% payment to be with held in the bill for effective performance of the contractor and it will be refunded after completion of one year performance period.

3. No communication from any tenderer adding to / altering or explaining any terms of the tender will be considered after the opening of the tenders by the Registrar. If any tenderer withdraws his tender after the tenders are opened, his tender deposit it is liable to be forfeited. Supplies should be effected within the delivery period quoted with tendered rates, failing which the tender deposit of the successful tenderer will be forfeited, and the order placed also cancelled at their risk and cost. Conditional offer such as subject to prior sales etc. are liable to be summarily rejected.

4. The Registrar, Alagappa University, Karaikudi reserves **the right to accept or reject** any tender either partly or wholly without assigning any reason therefore.

5. The quantities mentioned are only tentative and are liable to be added or subtracted.

6. No suit or any proceedings in regard to any matter arising in any respect under this contract shall be instituted in any Court save in the District Munsif Court or Sub Court at Sivaganga. It is agreed that no other Court shall have jurisdiction to entertain any suit or proceedings even through part of the cause of the action raised within their jurisdiction. In case of any part of cause of action arises within the jurisdiction of any of the Court in Tamil Nadu and not in the Court of Sivaganga then it is agreed to between the parties that such suits or proceedings shall be instituted in a court within Tamil Nadu and no other court outside Tamil Nadu shall have jurisdiction even though any part of the cause of action might arise within the jurisdiction of such courts.

7. The successful tenderer should arrange to supply the equipment as per terms of the accepted tender. Full payment with due deduction if any will be made to the supplier only after the receipt of the materials at site in good condition and after making necessary tests. Any loss or damages in transit should be made good by the supplier at free of cost. The advance payment will not be made under any circumstances. Defects in manufacture if any detected at any time should be rectified at free of cost.

8. The decision of the Registrar is final in respect of defective equipment. Quantities given in the tender are subject to alteration at the time of placing orders by the University.

9. The Successful tenderer should produce income-tax clearance certificate. In the case of partnership firms, this should be produced for each of the partners and the firm.

10. The tender should be valid for a period of 90 days from the date of opening.

11. Only parties who agree to abide by the above terms and conditions may send tenders and submission of tenders against this specification would signify such acceptance of the above terms and conditions by the tenderers.

12. Tenderers should specifically note that their tenders are liable to be summarily rejected if
    i. the rate quoted is not F.O.R. destination
    ii. complete technical details of the products as the case may be are not sent with the tender
    iii. validity period is not indicated
    iv. EMD is not remitted
13. Detailed Specifications of the equipment required are enclosed.

Note:
An EMD i.e. Rs.48,000/= through a Demand Draft drawn in favour of The Registrar, Alagappa Unviersity and payable at Karaikudi should accompany the tender.

Last date for receipt of tenders is upto 03.00 p.m. on 10.10.2012

REGISTRAR

Tender Eligibility Conditions:

The tender called for is a two bid tender and the technical & commercial bids are to be offered separately. The commercial bids of only those bidders who qualify technically will be opened

The intending bidders shall satisfy the following pre-qualification requirement.

1. The bidder must be ISO 9000 – 2008 certified for research and design of ICT based solutions apart from Product supply and delivery.

2. Bidder must be a registered SSI unit for Services in ICT technology.

3. The bidder company should have registered a turnover of at least Rs 1 crore per year in the immediate preceding three years (not inclusive of the turnover of associate companies) as per the audited accounts.

4. The bidder company should have registered net profit in the immediate preceding three years as per the audited accounts. Bidder should furnish duly approved copies of last 3 years Balance sheet and Profit and Loss accounts as proof of having met the above requirements.

5. Bidder should have local office, local service center within a radius of 100 kms. with full fledged training facilities for wireless networks for training our staff on regular basis. The technical evaluation team will inspect the premises to ascertain the availability of such facilities. Sufficient Proof with last one years' telephone bill from a Government Telecom organization in the bidders name specifying the location must be enclosed.

6. The bidder must have more than 5 years experience in supply, installation, and Technical support of Wireless Installation. Order copies and satisfactory certificates from Government customers to this effect need to be produced.

7. Bidders must have the experience of successful completion of at least one single project for a value of more than Rs.1 Crore in LAN / WAN/Wireless IP network infrastructure in any Government organization anywhere in India comprising supply, installation and commissioning.

8. The bidder must have completed minimum of 10 reputed PSU's / institutions / corporate installations of wireless of the same make in and around Tamil Nadu and running for more than one year and must have been executed by the successful bidder without any third party subcontract.

9. Preference to the bidder's who have already done wireless network to our university.

10. The bidder should be the authorized representative / dealer / distributor of the OEM. The bidder must have back-to-back support relation with the OEM whose products are offered by the bidder. The proof in support of the same must be enclosed along with the technical bid. Bids from agents without proper Authorization from the OEM for this particular tender / project shall be treated as non-responsive.

11. The OEM vendor must have Office, R&D and service center in India with 24 x 7 supports and it must be possible for any customization if required. Sufficient proof must be provided.

12. The Access Point, Access Point management controller and User Management Gateway Appliance should be from the same OEM and products from different OEM's will be summarily rejected.

13. The OEM should have certification of ISO 9001:2008 certification for its Quality Management Systems to cover design, manufacturing, sales and service of Wireless Access systems.

14. Offers should be enclosed with Product brochures of the make and model being quoted. Any deviations in technical specifications will be summarily rejected.

15. The detailed technical specifications for the products offered have to be complied with as per the tender and a written confirmation of compliance or deviation has to be categorically mentioned and if there is a compliance of a feature the same has to be backed by the specifications in the data sheet confirming the same. Tenders / Bids received with words like "As per your tender specifications" will be summarily rejected and their commercial offers will be returned back duly to them.

16. The bidder must quote for all the products as per the tender and must comply with all the tender specifications as per the tender.

17. The Technical and commercial details are to be submitted separately super-scribing "Cover 'A' - Technical details, Cover 'B' – Commercial details and Cover 'C' should contain EMD", Tender document fee, signed tender document, Supporting documents for the above eligibility criteria and Schedule with make and model details without pricing. All three covers are to be put together in one envelope and sealed properly, super-scribing the Tender Reference number.

18. If the bidder fails to furnish all the relevant documents / information as mentioned in the Tender Document, the Tender will be rejected and their Technical and Commercial document will be returned without opening the same.

19. Each bidder should submit authorization letter from the OEM. Only one OEM can be represented by one firm.

20. Transportation of materials to the execution sites are to be done by the contractor.

21. Rates to be quoted inclusive of all taxes, charges, etc. No extra cost is payable over and above the accepted rates.

22. All the materials make and model should be from reputed brand and must comply with all Technical compliance as per the detailed Technical Specifications attached.

23. All the materials should cover on site warranty for a period of 12 (Twelve) months from the date of commissioning. During warranty period, the contractor has the responsibility to replace/repair the defective equipments / parts etc. for proper functioning of the whole system round the clock.

24. In addition to the quotes pertaining to column No.23, the rates may be quoted including extended warranty for a further period of three years.

25. Annual maintenance charges after the expiry of warranty / extended warranty should be quoted (for 5th and 6th year).

Price Schedule(Rates should be filled in figures and words)

Sr. No	Brief Specification	Make & Model	Unit Price	Quantity	Price
1	Supply of Indoor Extended range indoor AP with 11n MIMO-Enterprise Class, High Gain antenna managed product with PoE Enabled as per the Technical specification **Annexure 1**. IPV6 network compatible.			118	
2	Supply of outdoor Extended Range multi-client device with integrated antenna, AP 2.4 GHz with SMART MIMO (802.11n) as per the Technical Specification **Annexure 2.** IPV6 network compatible.			62	
3	Supply of outdoor Extended Range multi-client device with integrated antenna, AP 5 GHz with SMART MIMO (802.11n) as per the Technical Specification **Annexure 3.** IPV6 network compatible.			16	
4	Supply of Hardware based Access Point management controller appliance to support up to 250 AP, 2500 user as per the Technical specification **Annexure 4.** IPV6 network compatible and IPS/VLAN			1	
5	Supply of Hardware based User Management Gateway Appliance for 2000-3000 users as per the Technical specification **Annexure 5.** Included software licence			1	
6	Supply of 12U Rack with Power Manager, Cooling fan as per the Technical Specification **Annexure 6.**			1	
7	Supply of 8 port 10/100/1000Mbps L2 manageable Network Switch as per the Technical Specification **Annexure 7.**			5	
8	Supply of 10/100/1000Mbps Media converter as per the Technical Specification **Annexure 8.**			10	
9	Supply of 10 feet 1" GI Pole as per the Technical specification **Annexure 9**.			27	

10	Supply of CAT 6 UTP cable box.			8	
11	Supply of RJ 45 Jack box.			8	
12	Supply of Conduit for UTP Cables.			2640 m	
13	Supply of 8 Port unmanageable Network Switch as per the Technical specification **Annexure 10**.			60	
14	Installation (Location details vide **Annexure 11**), Testing, Commissioning and Training of above equipments.			LS	
15	Rate per Year for three years extended Warrenty				
16	AMC rate per year for 5th and 6th year(After extended warrenty.)				
				Total	

Technical Schedule

Sr. No.	Brief Specification	Quantity
1	Supply of Indoor Extended range indoor AP with 802.11n MIMO-Enterprise Class, High Gain antenna managed product with PoE Enabled as per the Technical specification **Annexure 1**. IPV6 network compatible.	118
2	Supply of Outdoor Extended Range multi-client device with integrated antenna, AP 2.4 GHz with SMART MIMO (802.11n) as per the Technical Specification **Annexure 2**. IPV6 network compatible.	62
3	Supply of outdoor Extended Range multi-client device with integrated antenna, AP 5 GHz with SMART MIMO (802.11n) as per the Technical Specification **Annexure 3**. IPV6 network compatible.	16
4	Supply of Hardware based Access Point management controller appliance to support up to 250 AP, 2500 user as per the Technical specification **Annexure 4**. IPV6 network compatible & IPS/VLAN.	1
5	Supply of Hardware based User management Gateway appliance for 2000-3000 user as per the Technical specification **Annexure 5**. Included software licence.	1
6	Supply of 12U Rack with Power Manager, Cooling fan as per the Technical Specification **Annexure 6**.	1
7	Supply of 8 port 10/100/1000Mbps L2 manageable Network Switch as per the Technical Specification **Annexure 7**.	5
8	Supply of 10/100/1000Mbps single mode / Multi Mode Media converter as per the Technical Specification **Annexure 8**.	10
9	Supply of 10 feet 1" GI Pole as per the Technical specification **Annexure 9**.	27
10	Supply of CAT 6 UTP cable box	8
11	Supply of RJ 45 Jack box	8
12	Supply of Conduit for UTP Cables	2640 m
13	Supply of 8 Port unmanageable Network Switch as per the Technical specification **Annexure 10**.	60
14	Installation(Location Details **Annexure 11**), Testing, Commissioning and Training of above equipments.	LS

Bench mark for Bandwidth, security and throughput in accordance with International standard (IEEE standard for security for wireless).

Technical Specification:

Annexure 1. Indoor Access Point

Specifications	Brief Description	Compliance
Wireless Interface	Single radio, 802.11a/b/g/n, 2.4Ghz or 5Ghz	
Operating Frequency	2.412-2472 GHz and 5.150 GHz - 5.850 GHz-As per regulatory for Indoor AP's.	
Spatial Stream	AP Must support 2x2 MIMO.	
Configurable data rate	It should be possible to manually configure data rate on Wi-Fi Interface	
Channel bandwidth	5/10/20MHz & 20/40MHz	
WMM	Support for Wireless Multi Media	
Data rate	300 Mbps	
Adaptive modulation for Wi-Fi	BPSK, QPSK, 16QAM, 64QAM DSSS-CCK	
Uninterrupted Service	In case of Controller goes down, services should not affect and no compromise on security on the Edge.	
Stand Alone	Access Points should perform encryption / decryption on itself so as not to bottleneck the controller	
Operation Modes	Access Point should support the following Wireless Configuration: • Access point • Station • Station WDS • Access Point WDS • Repeater WDS • Station + Router • Access Point + Router	
Antenna	Access Point should have two external 5 dBi omni-directional, with minimum 1x2 MIMO operation	
Bandwidth supported with automatic Fallback	802.11b: 1, 2, 5.5, 11Mbps; 802.11g: 6,9,12,18,24,36,48, 54Mbps; 802.11a: 6,9,12,18,24,36,48, 54Mbps; 802.11n: MCS 0-15 up to 300Mbps	
network interface	1*10/100 Mbps Ethernet port Auto MDI/MDI-X support	
Management	Web-based configuration and management tool	

VLAN	Access Point should compliant with 802.1q VLAN support	
Input Power	802.11n Access Point should be able to power up using standards 802.3af POE input, and at the same time operate in full MIMO mode.	
Transmit Power	Access Point should provide minimum 20dBm transmission power (limited as per govt regulation for indoor AP's).	
Power Flexibility	Should capable of Varying the power level	
Security	• 64/128 bit WEP data encryption • MAC address filtering • WPA-EAP • WPA2– EAP • WPA - PSK • WPA2-PSK • 802.1x	
User Authentication	AES, SSID Broadcast Disable function, multiple SSID	
SSID support	Should support minimum 4 BSSID	
Wireless Scanning	Access Point capable of supporting part time scanning - serving clients and scanning the RF spectrum at the same time	
Operating Temperature	Should support the operating temp 0° to 40° C and Humidity: 5 to 90% non-condensing.	
Certification	ISO 9001:2008 (QMS)	
Weather proofing:	All the mast and out door equipments should be rust proof and weather proof	

Annexure 2. Outdoor 2.4 GHz Access Point

Specifications	Brief Description	Compliance
Wi-Fi Standards supported	AP shall support 802.11 b/g/n standards	
Wi-Fi Channel Support	Wi-Fi shall support channel allocations as per the Indian Wi-Fi Standards (Vendor to provide datasheet)	
Radiation under regulation	CPE/AP should radiate as per the EIRP limit set for the Indian Standards	
Wireless Medium	802.11 b/ g/n: OFDM	
frequency	2412 - 24835 MHz	
Channel bandwidth	5/10/20MHz & 20/40MHz	

WMM	Support for Wireless Multi Media	
Transmit Power	Minimum Radio Tx Power of 23 dBm. Vendor to specify the Transmit Power per Channel	
Power Flexibility	Should be possible of Varying the power level	
uninterrupted Service	In case of Controller goes down, services should not affect and no compromise on security on the Edge.	
Stand Alone	Access Points can perform encryption / decryption on itself so as not to bottleneck the controller	
Antenna type	Integrated Antenna with 14dBi antenna gain, (customizable to external antenna)	
Adaptive modulation for Wi-Fi	BPSK, QPSK, 16QAM, 64QAM DSSS-CCK	
Data rate	Support up to 150 Mbps	
Configurable data rate	It should be possible to manually configure data rate on Wi-Fi Interface	
Security Encryption	WEP, WPA, WPA2, AEP, EAP, AES. 802.1X between CPE/AP, CPE/AP shall support 128 bit WEP, 4 default transmit keys, WPA2 Pass Phrase, 4 WEP Keys	
DHCP support	CPE/AP interface should support DHCP mode (DHCP IP transparently received from the Network DHCP Server)	
Operation Modes	• Access Point • Station • Access Point WDS • Station WDS • Repeater WDS • Station + Router • Access Point + Router	
Power Over Ethernet	Power over Ethernet to avoid electrical cable for aesthetic purpose. POE should comply to 802.3af	
Interface Types	1*10/100 Mbps Ethernet port Auto MDI/MDI-X support	
Certification	ISO 9001:2008 (QMS)	
Power Flexibility	Capable of Varying the power level	
Weather proofing:	All the mast and out door equipments should be rust proof and weather proof	
Enclosure	Should be IP67 standard	

Annexure 3. Outdoor 5 GHz Access Point

Specifications	Brief Description	Compliance
Wi-Fi Standards supported	AP shall support 802.11 a/n standards	
Wi-Fi Channel Support	Wi-Fi shall support channel allocations as per the Indian Wi-Fi Standards.	
Radiation under regulation	CPE/AP should radiate as per the EIRP limit set for the Indian Standards	
Wireless Medium	802.11 a/n: OFDM	
frequency	5.8 GHz	
Channel bandwidth	5/10/20MHz & 20/40MHz	
WMM	Support for Wireless Multi Media	
Transmit Power	Minimum Radio Tx Power of 23 dBm. Vendor to specify the Transmit Power per Channel	
Power Flexibility	Capable of Varying the power level	
Uninterrupted Service	In case of Controller goes down, services should not affect and no compromise on security on the Edge.	
Stand Alone	Access Points can perform encryption / decryption on itself so as not to bottleneck the controller	
Antenna type	Integrated Antenna with 16dBi antenna gain, (customizable to external antenna)	
Adaptive modulation for Wi-Fi	BPSK, QPSK, 16QAM, 64QAM DSSS-CCK	
Data rate	Bandwidth of 150Mbps	
Configurable data rate	It should be possible to manually configure data rate on Wi-Fi Interface	
Security Encryption	WEP, WPA, WPA2, AEP, EAP, AES. 802.1X between CPE/AP, CPE/AP shall support 128 bit WEP, 4 default transmit keys, WPA2 Pass Phrase, 4 WEP Keys	
VLAN	802.1q VLAN support	
DHCP support	CPE/AP interface should support DHCP mode (DHCP IP transparently received from the Network DHCP Server)	
Operation Modes	• Access Point • Station • Access Point WDS • Station WDS • Repeater WDS	

	• Station + Router • Access Point + Router	
Power Over Ethernet	Power over Ethernet to avoid electrical cable for aesthetic purpose. POE should comply to 802.3af	
Interface Types	1*10/100 Mbps Ethernet port Auto MDI/MDI-X support	
Certification	ISO 9001:2008 (QMS)	
Enclosure	Should be IP67 standard	
Weather proofing	All the mast and out door equipments should be rust proof and weather proof	

Annexure 4. Access Point Management Controller Appliance

Specifications	Brief Description	Compliance
1.Hardware Specification		
Form factor	Hardware : Rack-mounting (standard 19" 1U configuration)	
Console / USB	1 RJ-45 / USB	
Design	Solution Design should be based on Centralized WLAN controller, Single point management for configurations for WLAN, Security, Access Control, 802.11n access points.	
Gigabit Ethernet on Chassis	Should have minimum 4 GE port	
Storage	Should have minimum 4GB storage to enable AP logs	
	Controller should comply with layer 2 features	
Ports	The wireless controller should have management ports (RJ-45+USB+ethernet).	
Power & Environment	Power supply unit should comply to Indian power supply standards. Ambient Operating 0~40 C/32~104 F Ambient Storage -20~70 C/-4~158 F	
Access Point Management		
	The wireless controller should function as an authenticator for all AP's	
	Should have adequate features to ensure the ability to auto discover AP's	
	Access Point Mapping	

	Redundancy/ Fault Management of Access Points.	
	Rogue Access Point Detection	
	RF Configuration for Access Points in both remote and local	
	Should be enable with Multi administration logging support	
	Remote Access Point Management	
	Each wireless controller should be able to accommodate minimum 254 Access Points	
	Provides centralized management via HTTP	
	Should support for Dual band configuration	
	Centralized Firmware Upgrade mechanism for All Managed Access Points	
	Access Point Logs	
	Email Notification	
	Grouping	
	Radius Authentication 802.1x	
	MAC Authentication	
	Controller should able to detect the fault Access Points	
Subscriber Management		
	User Authentication	
	User Profile Management	
	User Logs	
	Bandwidth Settings	

Annexure 5. User Management Gateway Appliance

Specifications	Brief Description	Compliance
1.Hardware Specification		
Form factor	Hardware : rack-mounting (standard 19" 1U configuration)	
Console / USB	1xRJ-45 / 2 USB	
Gigabit Ethernet on Chassis	Should have minimum 4 GE port	
2.Subscriber Management		
	Should support for 2000 subscribers	
	No additional client software to be installed	
	Should able to Control/Restrict subscribers based on - Static/Dynamic IP,	

	Ethernet MAC, Web-page, Portal Redirection and VLAN(802.1q)	
	Should Manage individual subscribers or create User Groups that share bandwidth and rate plans	
	Should able to Capture demographic details for all subscribers	
	Create Invoice for Billing	
3.Bandwidth & QoS Management		
	Should Support for multiple bandwidth management techniques – HTB, CBQ	
	should able to Associate subscribers to any one of multiple uplink gateways, pre-allocate contention ratio (1:1, 1:2, 1:4, etc.), groups/pools and accurately manage upstream and downstream bandwidth	
	Prioritization of VIP traffic	
	Time based Variable bandwidth allocation	
	Data usage based Variable bandwidth allocation	
	Bandwidth quota allocation based on individual and shared basis	
	Protocol Controlling & Blocking (For example rate limit or drop ICMP to protect from DDoS and other Worm attacks)	
	Should Block P2P traffic	
	Committed (CIR) bandwidth allocation	
	Brustable (CBR) bandwidth allocation	
4.Authentication, Authorization & Accounting (AAA)		
	Should be capable of Multi-mode authentication – use IP, MAC, Username and VLAN in any combination to acquire subscribers.	
	Should Support for secure (SSL) web page login via portal redirection	
	should Support for PPPoE dialer and RADIUS authentication	
	Support for 802.1x, EAP and AAA for third party access device integration e.g. wireless AP, DSLAM, Ethernet switch	
	CLI support for dial-RAS environment	
	Automatic disconnection based on subscriber's policy/plan/package	
	Should support 802.1X via radius	

5.Package & Rate Plan Management		
	Should define unlimited packages & plans;	
	Prepaid and post paid subscription should be created for usage (data transfer) and time-based plans	
	Support for Hourly, daily, monthly packages	
	Should Restrict based on time schedules, day of week and even special days such as holidays	
	Night package & Discounts support	
	"Pulse rate" support for pre-rating bytes & time usage. Cater to peak hours by tweaking pulse rate only and not changing various other plans	
	Complete support for "leased line" packages	
	Should define multiple packages against the same subscriber account	
	Traffic based rate plan support – 128kbps for Internet, 64kbps for VoIP, 256kbps for local Video Movie server, etc. All rate plans should be delivered simultaneously with no changes at subscriber PC	
	Online registration and renewal	
	Auto renewal with assigned Quota and time	
	should have Toll-Free package support that can be used to provide "free" access to certain sites or hosts	
6.Customer Web Self-Care		
	Customer Web Self-care Password change facility	
	Subscription and Renewal through PINs	
	View of bandwidth usage graphs for users	
	Diagnostic Tool-Network connectivity health checking tool	
	Usage Log based on time and data transfer	
7.Administration		
	Role-based administration	
	Multiple administrator and user levels	
	Upgrades and changes via Web UI	
	Auto backup of database, logs and usage	
	DDNS – support for services offered by following	

8.Firewall		
	Basic firewall	
	Content filtering	
	Database of Web Surf logs	
	Download/Reports of Web Surf logs	
	VLAN management	
9.Multiple Gateway Load Balancing		
	Multiple ISP links support	
	Premium customers should be diverted to high-speed internet links	
10.Routing		
	Static Routing	
	One to One or Many to One Source NATing	
	VLAN management	
	Snatting	
11.Reports		
	URL History	
	Maximum Usage Report	
	Expired Subscription details	

Annexure 6. 12U Rack

Specifications	Brief Description	Compliance
Size	The rack shall be 12U size	
Locking	It shall have Secure locking front and back doors with spring loaded hinges	
Front door	The Front door shall have a full length smoked shatterproof glass panel with decorative accent trim, and integrated locking mechanism.	
Mounting rails	It should have adjustable front and back mounting rails to allow flexible configuration.	
Cooling system	It shall have the dedicated space for top mounted cooling system.	
Cable Entry points	It shall support multiple cable entry points to provide maximum versatility. There are pre-punched holes on the top of the cabinet, rear side of the bottom and top cases.	

Glass Caster	The Front door shall have toughened glass Caster with foot operated lock	

Annexure 7. L2 Manage able 8 port Ethernet Switch

Specification	Brief Description	Compliance
Interface	• 8 port 10/100 Base-Tx + 2 port 1000 Base-T/SFP Ports. • 1 U Rack/Desktop Mount, RS 232 Console Port	
L2 Features	• Asymmetric VLAN/ISM VLAN. • Spanning Tree Protocols: 802.1D, 802.1w, 802.1s. • Link Aggregation: 802.1AX, 802.3ad • Double VLAN.	
Security	SSH/SSL, ACL, IP-MAC Port Binding, 802.1x Port/Host based Access Control, Guest VLAN.	
Monitoring and control	• Should support Port Mirroring. • Should support Broadcast/Unicast/Multicast storm control	
Management	• Should have SNMP v1,v2,v3 & RMON v1,v2 • Single IP Management • Network Load Balancing • LLDP.	
Performance	• Should have 5.6 Gpbs switching capacity. • 8K MAC Address Entries. • 16 MB Flash Memory. • Packet forwarding rate should be 4.2 Mbps. • Packet Buffer should be min 384 KB.	

Annexure 8. Media Converter

Specification	Brief Description	Compliance

Port Density	• Should have one 10/100/1000 Base-T Copper Port • Should have one 1000 Base-X Gigabit Fibre Port. • SC type optical connector • Single mode / Multi mode	
Standards	Should support following IEEE standard compliance. • IEEE 802.3ab • IEEE 802.3u • IEEE 802.3z for 1000 Base-T & 1000 Base-X interfaces.	
Configuration	The Ethernet configuration should have Frame support, Flow control, Admin status, Auto negotiation.	
Management	• SNMP v1, v2 support. • SNMP Trap Management for error, Link status, Firmware changes & Power failure.	

Annexure 9. GI Pole

Specifications	Brief Description	Compliance
	The Pole should be fabricated using high grade raw material and latest technology in compliance with set industry norms.	
	The Pole should have 25.4 mm thickness	
	The Pole should have 3.3 meter height	
	The Pole should be Corrosive resistant.	
	The Pole should have longer service life and easy to maintain.	
	The pole should be durable and high strength and low maintenance.	

Annexure 10. Unmanageable 8 port Switch

Specifications	Brief Description	Compliance
	The switch shall be 8 port 10/100 Mbps Fast Ethernet ports	
	It should support Auto MDI/MDIX crossover for all ports	
	It shall have the Store-and-forward switching scheme	

	It should support both Full/half-duplex for Ethernet/Fast Ethernet speeds	
	It shall have the IEEE 802.3x Flow Control feature.	
	It should support Plug-and-play installation	
	The switching fabric should be 1.6 Gbps	
	It should support CSMA/CD protocol	
	It shall have the following feature Ethernet at 10 Mbps (half duplex) and 20 Mbps (full duplex) and in Fast Ethernet 100 Mbps (half duplex) and 200 Mbps (full duplex)	
	It should support Star Topology	
	The Media Interface Exchange shall be Auto MDI/MDIX adjustment for all ports	
	It should have the LED Indicators of Link/Activity Per port and Power Per device	
	The Transmission Method shall be Store-and-forward mechanism	
	It should support MAC Address Table of 2K entries per device with MAC Address Learning and Automatic update	
	The Packet Filtering/Forwarding Rates Ethernet shall be 14,880 pps per port And in Fast Ethernet it should be 148,800 pps per port	
	It should support RAM Buffer of 57 Kbytes per device	
	The Heat Dissipation during Power On (Standby) shall be AC input 4.774 BTU/h And Maximum AC input: 17.732 BTU/	

Annexure 11. Location Details

Science Block

- Main Building (Six floors)
- Guest House
- Ladies Hostel
- Boys hostel

University Campus

- Library - 2 floors
- Main Building - 2 floors
- Auditorium
- D.D. Exam - 3floors
- Convocation
- Tamil
- Economics
- Women Studies - 3 Floors
- Eng & Foreign
- Industrial Chemistry - 2 floors
- Post Graduate Block
- Distance Education - 2 floors

Ramanujam Dept

- Ramanujam Maths
- Director Of D.Education
- Ladies Hostel

P.ED

- Department Of Management
- Physical Education
- Boys hostel
- Girls hostel

Others

- Vishalakshi Boys Hostel
- Bio-Technology
- VC House
- Registrar Home
- International Guest House
- Faculty Guest House
- B.ED Ladies Hostel
- College Of Education
- Old Auditorium
- Staff Quarters